Assignment - 3

Course: MST501-1: Python Programming for Statistics

Done by:

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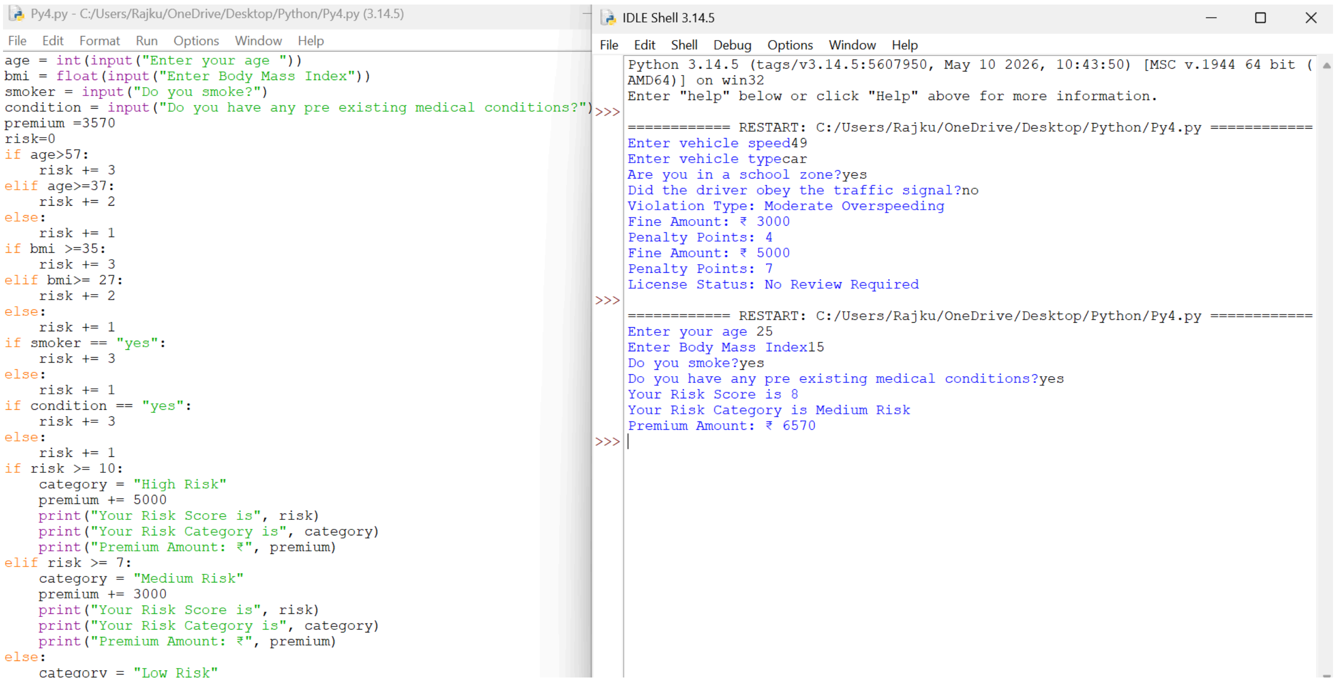
1.Develop a Python program that determines a
customer's health insurance premium category based on age, BMI, smoking status, and pre-existing medical conditions. Use if and if-else statements to classify risk levels and calculate the applicable premium amount.

Ans-
```
age = int(input("Enter your age "))
bmi = float(input("Enter Body Mass Index"))
smoker = input("Do you smoke?")
condition = input("Do you have any pre existing medical conditions?")
premium =3570
risk=0
if age>57:
    risk += 3
elif age>=37:
    risk += 2
else:
    risk += 1
if bmi >=35:
    risk += 3
elif bmi>= 27:
    risk += 2
else:
    risk += 1
if smoker == "yes":
    risk += 3
else:
    risk += 1
if condition == "yes":
    risk += 3
else:
    risk += 1
if risk >= 10:
    category = "High Risk"
    premium += 5000
    print("Your Risk Score is", risk)
    print("Your Risk Category is", category)
    print("Premium Amount: ₹", premium)
elif risk >= 7:
    category = "Medium Risk"
    premium += 3000
    print("Your Risk Score is", risk)
    print("Your Risk Category is", category)
    print("Premium Amount: ₹", premium)
else:
```

    category = "Low Risk"
    premium += 1000
    print("Your Risk Score is", risk)
    print("Your Risk Category is", category)
    print("Premium Amount: ₹", premium)




```
age = int(input("Enter your age "))
bmi = float(input("Enter Body Mass Index"))
smoker = input("Do you smoke?")
condition = input("Do you have any pre existing medical conditions?")
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risk=0
if age>57:
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    risk += 2
else:
    risk += 1
if bmi >=35:
    risk += 3
elif bmi>= 27:
    risk += 2
else:
    risk += 1
if smoker == "yes":
    risk += 3
else:
    risk += 1
if condition == "yes":
    risk += 3
else:
    risk += 1
if risk >= 10:
    category = "High Risk"
    premium += 5000
    print("Your Risk Score is", risk)
    print("Your Risk Category is", category)
    print("Premium Amount: ₹", premium)
elif risk >= 7:
    category = "Medium Risk"
    premium += 3000
    print("Your Risk Score is", risk)
    print("Your Risk Category is", category)
    print("Premium Amount: ₹", premium)
else:
    category = "Low Risk"
```



```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: C:/Users/Rajku/OneDrive/Desktop/Python/Py4.py ============
Enter vehicle speed49
Enter vehicle typecar
Are you in a school zone?yes
Did the driver obey the traffic signal?no
Violation Type: Moderate Overspeeding
Fine Amount: ₹ 3000
Penalty Points: 4
Fine Amount: ₹ 5000
Penalty Points: 7
License Status: No Review Required
>>>
============ RESTART: C:/Users/Rajku/OneDrive/Desktop/Python/Py4.py ============
Enter your age 25
Enter Body Mass Index15
Do you smoke?yes
Do you have any pre existing medical conditions?yes
Your Risk Score is 8
Your Risk Category is Medium Risk
Premium Amount: ₹ 6570
>>>
```

2. Create a Python program that analyses a driver's
speed, vehicle type, school-zone status, and traffic signal compliance.
Use if-elif-else statements to determine the type of violation, calculate
fines, assign penalty points, and identify whether the driver's license
should be flagged for review.

Ans-speed =int(input("Enter vehicle speed"))

vehicle = input("Enter vehicle type")

sclzone= input("Are you in a school zone?")

signal = input("Did the driver obey the traffic signal?")

fine=0

points=0

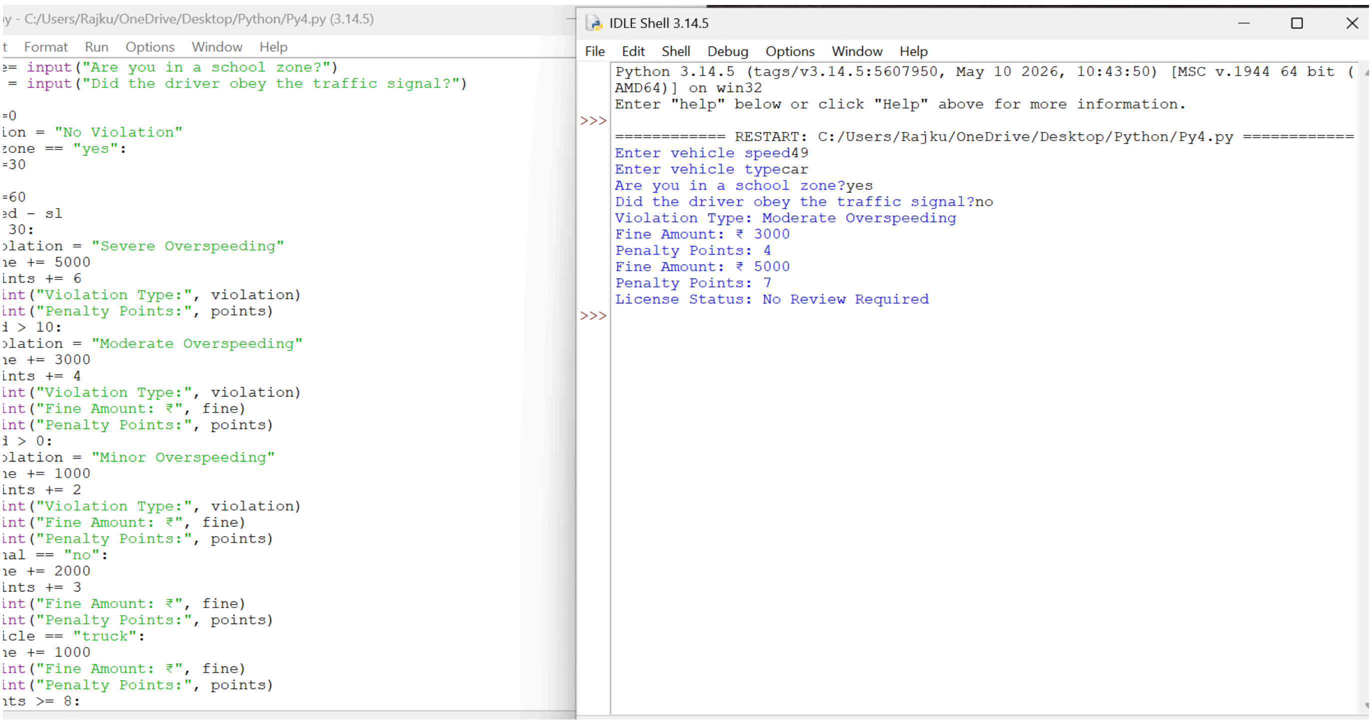
violation = "No Violation"

if sclzone == "yes":

    sl=30

```
else:
    sl=60
sd=speed - sl
if sd> 30:
    violation = "Severe Overspeeding"
    fine += 5000
    points += 6
    print("Violation Type:", violation)
    print("Penalty Points:", points)
elif sd > 10:
    violation = "Moderate Overspeeding"
    fine += 3000
    points += 4
    print("Violation Type:", violation)
    print("Fine Amount: ₹", fine)
    print("Penalty Points:", points)
elif sd > 0:
    violation = "Minor Overspeeding"
    fine += 1000
    points += 2
    print("Violation Type:", violation)
    print("Fine Amount: ₹", fine)
    print("Penalty Points:", points)
if signal == "no":
    fine += 2000
```

```
    points += 3
    print("Fine Amount: ₹", fine)
    print("Penalty Points:", points)
if vehicle == "truck":
    fine += 1000
    print("Fine Amount: ₹", fine)
    print("Penalty Points:", points)
if points >= 8:
    review = "License Flagged for Review"
    print("License Status:", review)
else:
    review = "No Review Required"
    print("License Status:", review)
```



3 Design a Python program for a citizen service
portal where users select services such as passport renewal, driving
license renewal, tax payment, utility bill payment, or grievance
submission. Use match-case to route requests to the appropriate service workflow.
Within each case, use nested if-elif-else statements to validate
eligibility, calculate fees, apply discounts or penalties, and generate a
final service summary.

Ans-

```
service=int(input("Select a service"))

fee = 0

match service:

    case 1:

        age = int(input("Enter your age"))

        expired = input("Has passport expired?")

        fee = 2000

        if expired == "yes":

            fee += 500

            summary = "Late renewal penalty applied"

            print("Total Fee", fee)

            print("Details", summary)

        elif age >= 60:

            fee -= 300

            summary = "Senior citizen discount applied"

            print("Total Fee", fee)

            print("Details", summary)

        else:

            summary = "Normal passport renewal"

            print("Total Fee", fee)
```

```
        print("Details", summary)
case 2:
    age = int(input("Enter age"))
    expired = input("Has license expired?").lower()
    fee = 1000
    if expired == "yes":
        fee += 300
        summary = "Late renewal penalty applied"
        print("Total Fee", fee)
        print("Details", summary)
    elif age >= 60:
        fee -= 200
        summary = "Senior citizen discount applied"
        print("Total Fee", fee)
        print("Details", summary)
    else:
        summary = "Normal license renewal"
        print("Total Fee", fee)
        print("Details", summary)
case 3:
    income = float(input("Enter annual income: "))
    if income < 300000:
        fee = 0
        summary = "No tax applicable"
        print("Total Fee", fee)
```

```
        print("Details", summary)
    elif income <= 700000:
        fee = income * 0.05
        summary = "5% tax applied"
        print("Total Fee", fee)
        print("Details", summary)
    else:
        fee = income * 0.10
        summary = "10% tax applied"
        print("Total Fee", fee)
case 4:
    bill = float(input("Enter bill amount"))
    days = int(input("Enter delayed payment days"))
    fee = bill
    if days > 30:
        fee += 500
        summary = "Late payment penalty applied"
        print("Total Fee", fee)
        print("Details", summary)
    elif days > 10:
        fee += 200
        summary = "Small penalty applied"
        print("Total Fee", fee)
        print("Details", summary)
    else:
```

```
        fee -= bill * 0.10
        summary = "10% early payment discount"
        print("Total Fee", fee)
        print("Details", summary)
case 5:
    priority = input("Enter priority")
    fee = 0
    if priority == "high":
        summary = "Priority grievance submitted"
        print("Details", summary)
    elif priority == "medium":
        summary = "Standard grievance submitted"
        print("Details", summary)
    else:
        summary = "Normal grievance submitted"
        print("Details", summary)
case _:
    print("Invalid service selected")
```

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```python
rvice=int(input("Select a service"))
e = 0
tch service:
  case 1:
      age = int(input("Enter age: "))
      expired = input("Has passport expired? (yes/no): ").lower()
      fee = 2000
      if expired == "yes":
          fee += 500
          summary = "Late renewal penalty applied"
          print("Total Fee", fee)
          print("Details", summary)
      elif age >= 60:
          fee -= 300
          summary = "Senior citizen discount applied"
          print("Total Fee", fee)
          print("Details", summary)
      else:
          summary = "Normal passport renewal"
          print("Total Fee", fee)
          print("Details", summary)
  case 2:
      age = int(input("Enter age"))
      expired = input("Has license expired? (yes/no): ").lower()
      fee = 1000
      if expired == "yes":
          fee += 300
          summary = "Late renewal penalty applied"
          print("Total Fee", fee)
          print("Details", summary)
      elif age >= 60:
          fee -= 200
          summary = "Senior citizen discount applied"
          print("Total Fee", fee)
          print("Details", summary)
      else:
          summary = "Normal license renewal"
          print("Total Fee", fee)
          print("Details", summary)
  case 3:
```

Ln: 6

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```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py4.py ===========
Select a service2
Enter age25
Has license expired? (yes/no): yes
Total Fee 1300
Details Late renewal penalty applied
>>>
```

Ln: 10